Nguyễn Đình Thiện

AI Engineer

035 669 4115 workthiennd@gmail.com https://github.com/thi3nnq Hanoi, Vietnam

OBJECTIVE

To leverage advanced Natural Language Processing techniques to develop robust, scalable solutions that extract valuable insights from unstructured text data, enhance user experiences through intelligent language understanding, and drive innovation in machine learning-driven language applications.

EDUCATION

Hanoi University of Science and Technology Sep 2018 - 2023

Major: HEDSPI Program (IT-E6)

CPA: 3.26/4

WORK EXPERIENCE

DNSE Securities and Outsourcing June 2024 - Now

AI Engineer

Project Description:

1. Ensa - Virtual stock trading assistant
    a. Building Multi-Agent System
    b. Build a database to store stock information.
2. Web Agent Autoppia
    a. Building Web Automation Agent for Shopping, Food Delivery Domain
    b. Training LLMs for information retrieval and web automation tasks
    c. Build a vector database to extract product information.
    d. Deploying LLMs

Tech Stack:

LLMs, LangGraph, LangChain, FastAPI, vLLM, LoRA, Milvus, MongoDB

ICOMM Media & Tech., Jsc July 2023 - May 2024

AI Engineer

Project Description:

1. Building Large Language Models for Vietnamese
    a. Building high-quality instruction datasets
    b. Pretraining and Instruction Fine-Tuning of the Llama2 (7B version) and TinyLlama (1.1B) Model for Vietnamese
    c. Achieving 40 Point Avg on VMLU (not-submitted)
    d. Quantizing model
    e. Deploying model
2. Topic-news Clustering
    a. Training and deploying translation model (zh-vi, en-vi)
    b. Using a hierarchical clustering algorithm and a dense retrieval model for Topic Clustering
3. Conversational Chatbot
    a. Developing pipeline for chatbot flows using Retrieval Augmented Generation
    b. Building retrieval modules and generator modules for medicine and banking domains

Technologies used:

BERT, BART, Elasticsearch, Docker, LLMs, Ctranslate2, GPT-Q, FastAPI, BGE-m3, Weaviate, LoRA, QLoRA

FTECH Feb 2022 - Mar 2023

NLP Engineer

Project Description: - Spelling Correction for Chitchat Domain
- Integrate ChatGPT into the website fqa.vn
- Paraphrase Generation to augment Chatbot Response Data - Intent Classification for Chatbot - Building Chatbot on Tiktok, and Facebook to answer customers and playersTechnologies used:PyTorch, Transformers, Rasa, scikit-learn, Docker, MongoDB, ChatGPT API, FastAPI, Triton Inference Server

OTHER PROJECT

DataScience Lab And PennyPilots Team -

- Building mongingon web-app (mongingon.site)

Description:
Create an app that allows users to choose trending delicacies and create each person's personal food map. The application uses NLP techniques to rate restaurants based on comments on social networking platforms.

- Building a chatbot for an e-commerce platform

Description:
Create a chatbot that can advise users about items such as clothes and shoes from an online shop.

- VLSP2022 - Vietnamese Abstractive multi-document summarization

Description:
Build LongBartPho model for multi-document summarization

- Zalo AI Challenge 2022 - E2E Quenstion Answering

Responsibilities:
- Handle noisy text data and augment data
- Build ElasticSearch and fine-tune MRC, CrossEncoder model

CERTIFICATIONS

Coursera's Machine Learning Certificate	2020
Coursera's Deep Learning Specialization Certificate	2021

HONORS & AWARDS

3rd place physics violympic in the province	2016-2017
1st place in Sun*'s Machine Learning Class	2020

SKILLS

Language::	English, Japanese
Mathematical skills::	Linear Algebra, Calculus, Probability Theory
NLP Frameworks and Libraries	PyTorch, NumPy, scikit-learn, Transformers, Llama-index, langchain
Databases	ElasticSearch, Solr, MongoDB, Weaviate
Programing Language::	Python, C